Let x_{ij} be predictor j for subject i; $y_i = (t_i, \delta_i)$ the survival outcome, $i = 1, \ldots, n$.
Partition predictors into B ordered blocks according to priority $\boldsymbol{\pi} = (\pi_1, \ldots, \pi_B)$.

Choose elastic net mixing parameter $\alpha \in [0, 1]$ and number of folds K.
Construct a decreasing regularisation path $\lambda_1 > \cdots > \lambda_m$.
Split $\{1, \ldots, n\}$ into folds $\{S_k\}_{k=1}^{K}$ and initialise $\hat{\eta}_{0,i}(\boldsymbol{\pi})_{\mathrm{CV}} = 0$.
for $b = 1, \ldots, B$ in priority order **do**
  Let $g = \pi_b$ denote the active block with predictors $X^{(g)}$.
  for $\ell = 1, \ldots, m$ **do**
    For each fold k, fit the penalised Cox model on $S \setminus S_k$

$$\hat{\beta}^{(g)}_{-k}(\lambda_\ell) = \arg\min_{\beta^{(g)}} \left\{ -\ell^{S \setminus S_k}_{\mathrm{Cox}}\big(\hat{\eta}_{b-1}(\boldsymbol{\pi})_{\mathrm{CV}} + X^{(g)}\beta^{(g)}\big) + \lambda_\ell \sum_{j \in g} \big(\tfrac{1-\alpha}{2}\beta_j^2 + \alpha|\beta_j| \big) \right\}.$$

    Using observations in fold S_k, compute the cross-validated deviance $\mathrm{CV}_b(\lambda_\ell)$.
  end for
  Select the block-specific tuning parameter

$$\lambda_{b,\min} = \arg\min_{\lambda_\ell} \mathrm{CV}_b(\lambda_\ell).$$

  Update cross-validated offsets for all subjects

$$\hat{\eta}_{b,i}(\boldsymbol{\pi})_{\mathrm{CV}} = \hat{\eta}_{b-1,i}(\boldsymbol{\pi})_{\mathrm{CV}} + \sum_{j=1}^{p_g} x_{ij}^{(g)}\, \hat{\beta}^{(g)}_{-k(i),j}(\lambda_{b,\min}),$$

  where $k(i)$ is the fold containing subject i.
end for
Final linear predictor: $\hat{\eta}_{B,i}(\boldsymbol{\pi})_{\mathrm{CV}}$.
Estimate the hazard function for all subjects:

$$\hat{h}_i(t) = h_0(t) \exp\big(\hat{\eta}_{B,i}(\boldsymbol{\pi})_{\mathrm{CV}}\big), \; i = 1, \ldots, n.$$